

AP CALCULUS AB - UNIT 3

Guided Notes 3.2: Implicit Differentiation



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we differentiate relations in which y is defined implicitly as a function of x ?

- Differentiate relations in which y is defined implicitly as a function of x .
- Solve algebraically for dy/dx and evaluate it at points on an implicit curve.
- Determine tangent and normal lines, horizontal and vertical tangents, and second derivatives.
- Recognize when a derivative formula is inconclusive at a singular point.

Key Knowledge, Vocabulary, and Notation

Implicit relation

An equation involving both x and y without first solving explicitly for y .

Derivative of a y -term

Apply the Chain Rule: $\frac{d}{dx}(y^n) = ny^{n-1}y'$ and $\frac{d}{dx}(\sin y) = \cos y y'$.

Product containing y

$$\frac{d}{dx}(xy) = xy' + y.$$

Horizontal tangent

Usually numerator = 0 and denominator $\neq 0$ in a simplified formula for y' .

Vertical tangent

Usually denominator = 0 and numerator $\neq 0$.

Second derivative

Differentiate the first-derivative relation again and substitute known values of y' .

Concept Development and Strategy

1. Differentiate every term with respect to x and attach y' whenever the differentiated expression contains y .
2. Collect all terms containing y' on one side before factoring and dividing.
3. Verify that a stated point lies on the original curve before using it.
4. For a normal line, use the negative reciprocal of a finite nonzero tangent slope.
5. If both numerator and denominator of the derivative formula vanish, return to the original relation; the formula is inconclusive.

Worked Example: Circle and tangent line

For $x^2 + y^2 = 25$, find y' and the tangent line at $(3, 4)$.

Answer and Reasoning

$$2x + 2yy' = 0, \text{ so } y' = -x/y. \text{ At } (3, 4), m = -3/4 \text{ and } y - 4 = -\frac{3}{4}(x - 3).$$

Worked Example: Product and trigonometric y -terms

For $x^2y + \sin y = 4$, find y' .

Answer and Reasoning

$$2xy + x^2y' + \cos y y' = 0, \text{ so } y' = -\frac{2xy}{x^2 + \cos y}.$$

Worked Example: Second derivative

For $x^2 + y^2 = 25$, find y'' at $(3, 4)$.

Answer and Reasoning

Differentiate again: $2 + 2(y')^2 + 2yy'' = 0$. With $y' = -3/4$, $y'' = -25/64$.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. For $x + y = 7$, find y' .

2. For $x^2 + xy + y^2 = 7$, find y' .

3. Write the tangent line to $x^2 + xy = 6$ at $(2, 1)$.

4. For $e^{xy} = x + y$, solve for y' .

5. For $x^3 + y^3 = 6xy$, find y' .

6. Show that no value of c makes $x^2 + cxy + y^2 = 6$ pass through $(1, 1)$ with slope -2 .

7. At $(0, 0)$ on $y^2 = x^2(x + 1)$, use leading terms to identify candidate tangent slopes.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. If $x^2 + y^2 = 9$, then y' equals

(A) $-x/y$

(B) $-y/x$

(C) x/y

(D) y/x

2. The derivative of $\sin y$ with respect to x is

(A) $\cos y$

(B) $\cos y y'$

(C) $-\sin y y'$

(D) $\sin y y'$

3. For $x^2 + y^3 = 6$, at $x = 1$ and $y > 0$, the slope is closest to

(A) -0.23

(B) -0.39

(C) -0.57

(D) -0.82

Common Errors and AP Precision

- Mark every layer or every occurrence of y before differentiating; omitted inner derivatives are the most common source of error.
- For inverse derivatives, find the preimage first. For implicit relations, verify the point lies on the original curve.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Prove $x^2 + xy + y^2 = 7$ has no point satisfying both horizontal- and vertical-tangent conditions.

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